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# Securing the GSTN

We have a chat with Nitin Mishra, EVP Services, GSTN to understand the infrastructure and security measures in place for our country's Goods and Services Tax Network.

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**Q Was there a requirement of rearchitecting the existing taxation structure to implement GST or is it an addition of the existing infrastructure?**

**Mishra:** One of the most unique thing about GST is that there was absolutely no parallel to draw inspiration from. There was no precedence to GST, at least in India or abroad. Our GST is different and more complicated than any other GST that has existed given the multiple rates, interstate GST, etc. It was pretty unique for our country. We did have some data that we had analysed to generate estimates such as the number of invoices generated and number of payments done. However, all these estimates went were useless because the taxpayer behaviour completely changed. Erstwhile, during the service tax regime, you were only required to have 8 lakhs as the annual turnover, which then became 20

lakhs under GST. In the erstwhile regime, the taxpayers had monthly payments to make and filing were on a quarterly basis. Under GST, the taxpayers have to make payments and filings on a monthly basis. The way that indirect taxation worked in India completely changed.

Implementing GST was a ground-up work. When we started conceptualising GST, technology was kept on the forefront. All the processes, from payments to filing returns to assessments and then on to adjudications, etc. were all to be done on the system. So a lot of these processes were redefined to align with the way that the technology platform would interact with the stakeholders including the tax-payers and officers.

**Q Since you mentioned that the volume of transactions with our GST implementation is much more, could you elaborate more on the data centres and the supporting infrastructure that are in use?**

**Mishra:** We have our own data centres. With GST, the Government did not want to take any risks. It's a centralised system working out of four data centres. Effectively, we have a DC (Data Centre) as well as a near-DC. And we also have a DR (Disaster Recovery) and a near-DR. We are operating in a functionally active-active mode, so we also have a few applications running in our DR as well so we aren't wasting any compute bandwidth over there.

**Q Could you shed some light on the redundancies in place?**

**Mishra:** Each component in GST has a fallback mechanism in an active-active mode. The network layers is working across two data centres in DC as well as in DR. Plus, we also have a seamless integration or our DR where all data that gets copied, or created in our DC seamlessly flows over to a DR. Now to ensure zero data-loss there is a distance between our two data centres and they're supposed to be in two seismic zones. The DR is in Bangalore and the DC is in Delhi. Though we have a very thick pipe, the laws of physics don't change. We also included an innovation called Near Data Centre (NDC) in the design. This way, there is always two synchronous copies of every single data that is committed, available with us. So we always have two copies within two distinct buildings in the same campus of our data centre. In this way we have ensured in-process redundancy and in case of down-times or if anything untoward were to happen, we always have the option to go to our DR.

**Q What security programmes are in place to secure the system? And do you have a bug-bounty programme?**

**Mishra:** We have over-invested in our security apparatus, in terms of application as well as infrastructure. We are in a tiered application architecture. When I say tiered, I mean, we have an in-source protection mechanism and our ISP themselves provide DDOS protection. Then our outermost layer is protected through various hardware and software systems. For example, we have a CAS (Content Analysis System) and we have proxy-servers to isolate the public from private traffic. Then we have firewalls to ensure only desired and designated traffic comes in. Moreover we have application firewalls to identify any untoward attempt which may pass through the outer layer. So we have all sorts of infrastructure and software invested. Moreover, we have employed auditing software for our code before they're pushed into production.

We are being audited by internal and external auditors. We are audited by the CAG and STQC, our own internal teams as well as external agencies. So there are multiple layers of checks that happen. 